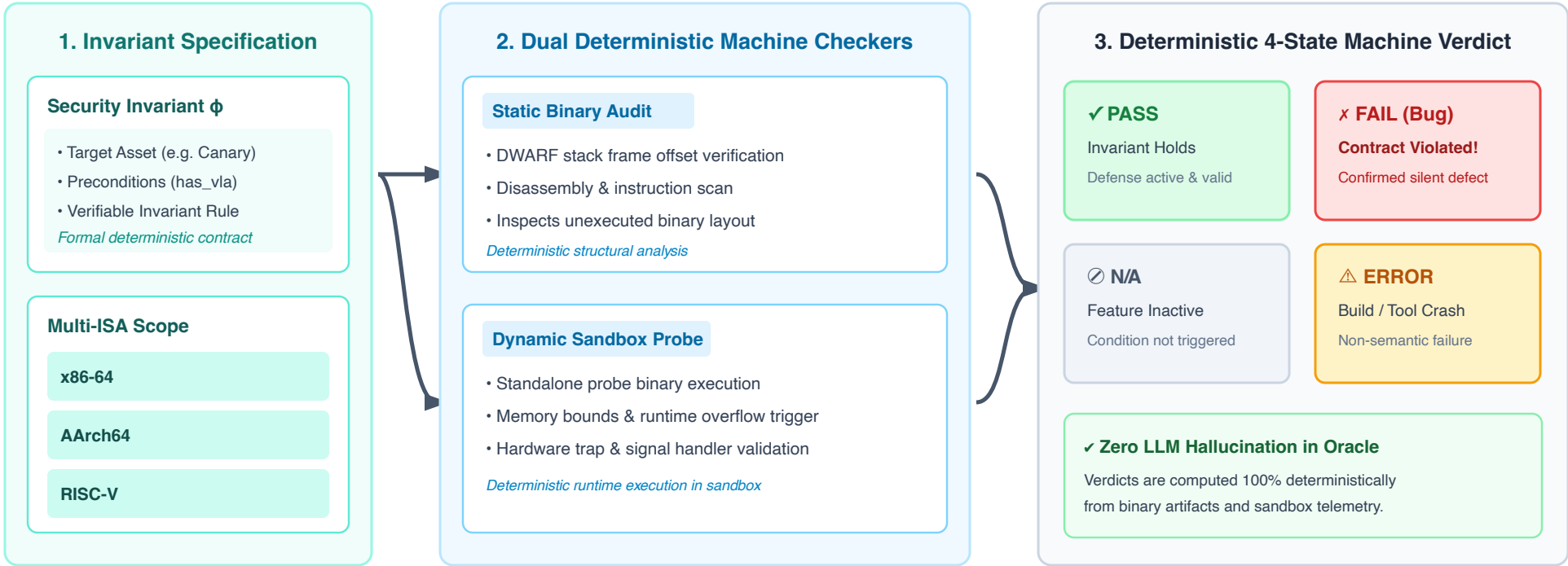

1. Invariant Specification
Security Invariant ϕ
• Target Asset (e.g. Canary)
• Preconditions (has_vla)
• Verifiable Invariant Rule
Formal deterministic contract
Multi-ISA Scope
x86-64
AArch64
RISC-V
2. Dual Deterministic Machine Checkers
Static Binary Audit
• DWARF stack frame offset verification
• Disassembly & instruction scan
• Inspects unexecuted binary layout
Deterministic structural analysis
Dynamic Sandbox Probe
• Standalone probe binary execution
• Memory bounds & runtime overflow trigger
• Hardware trap & signal handler validation
Deterministic runtime execution in sandbox
3. Deterministic 4-State Machine Verdict
✓ PASS
Invariant Holds
Defense active & valid
✗ FAIL (Bug)
Contract Violated!
Confirmed silent defect
⊘ N/A
Feature Inactive
Condition not triggered
⚠ ERROR
Build / Tool Crash
Non-semantic failure
✓ Zero LLM Hallucination in Oracle
Verdicts are computed 100% deterministically
from binary artifacts and sandbox telemetry.